ALLEN INSTITUTE for NEURAL DYNAMICS

NEW TECHNOLOGIES FOR MAPPING CENTIMETER SCALE TISSUES WITH NANOSCALE RESOLUTION

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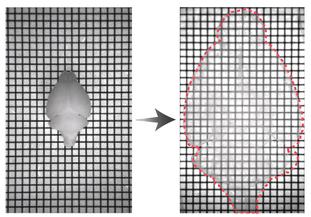
Also see POSTER 11

ABSTRACT

How do you image the large and the small at the same time? We developed new technologies for fast nanoscale imaging of centimeter-scale specimens - including whole mouse brains - with minimal tiling and no physical sectioning.

TISSUE EXPANSION

Imaging with diffraction-limited nanoscale resolution throughout centimeter-scale specimens requires tissue samples with negligible index of refraction variations. To achieve this goal, we developed a novel expansion method for centimeter-scale tissue samples. The tissue clarity and resolution improvement provided by this protocol reduces the technical burden on the downstream imaging systems.

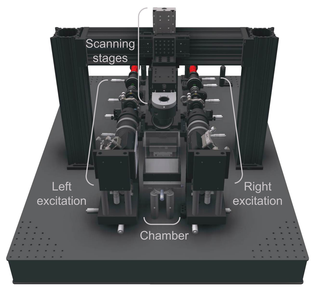


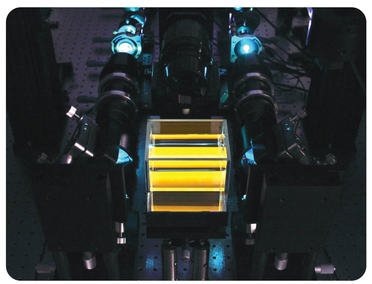
Before expansion | After expansion

1X EXA-SPIM SYSTEM

Imaging these physically enlarged tissues required a new imaging technology. Standard microscope objectives have limited working distances (< 1 mm) and fields of view (< 1 mm). Therefore, imaging large tissue volumes at high resolution typically requires physical sectioning and extensive image tiling, both of which severely complicate large-scale volumetric imaging. These technical constraints often limit current imaging technologies to imaging small volumes at high resolution (<1 μm) or large volumes at lower resolution (>1 μm). To address this need, we have developed a new type of micrscope - which we refer to as expansion-assisted selective plane illumination microscopy (ExA-SPIM). ExA-SPIM achieves an unrivaled combination of imaging resolution, speed, and volume by leveraging lenses and cameras developed for the semiconductor industry. To enable multi-scale imaging of specimens of various sizes and resolutions, we have developed 1X, 5X, and 10X versions of the system.

Lateral resolution	Axial resolution	Field of view	Imageable volume	Imaging speed
3.1 μm	9.0 μm	53.0 x 44.0 mm	90 x 100 x 35 mm	Up to 2 gigavoxels/sec

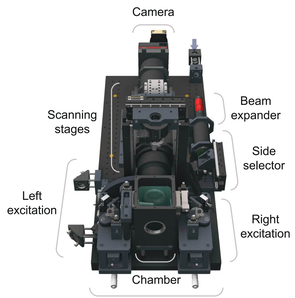


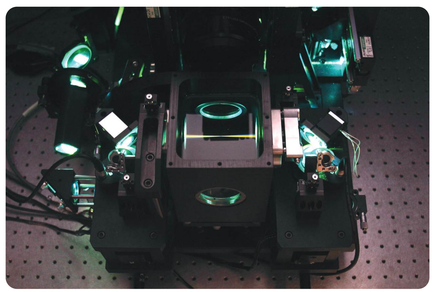


Base plate (900 x 1200 mm)

5X EXA-SPIM SYTEM

Lateral resolution	Axial resolution	Field of view	Imageable volume	Imaging speed
1.0 μm	3.0 μm	10.6 x 8.0 mm	90 x 100 x 35 mm	Up to 2 gigavoxels/sec

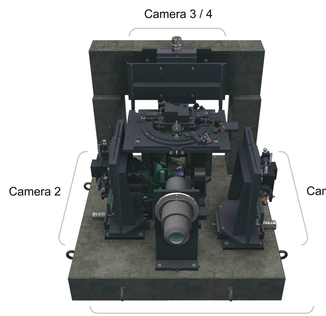


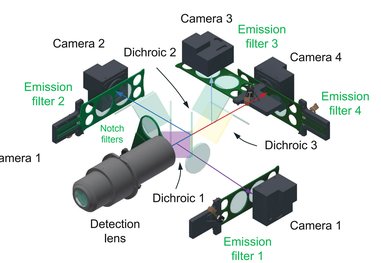


Aluminum base plate

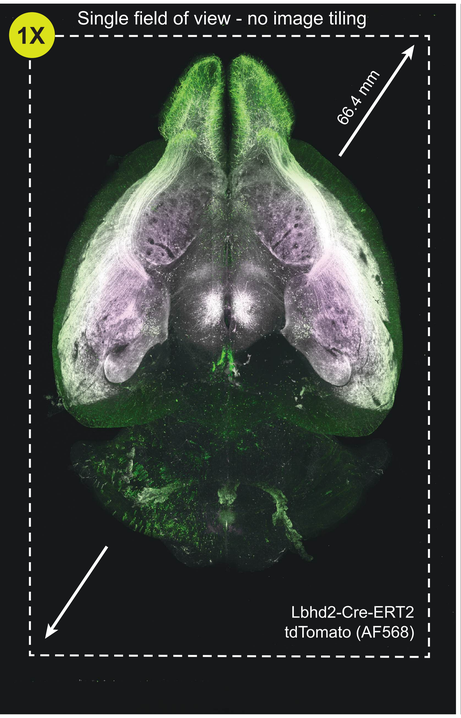
10X EXA-SPIM SYTEM

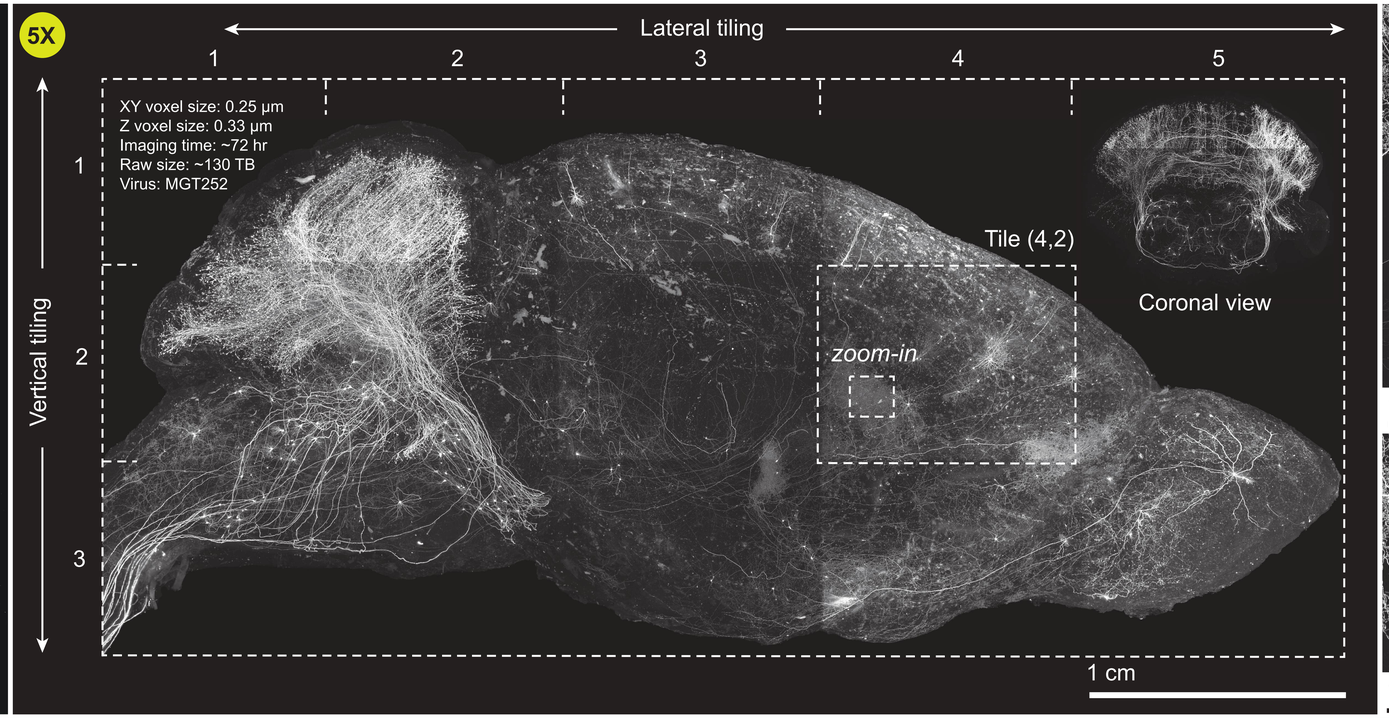
Lateral resolution	Axial resolution	Field of view	Imageable volume	Imaging speed
0.6 μm	0.8 μm	5.3 x 4.0 mm	90 x 100 x 35 mm	Up to 8 gigavoxels/sec

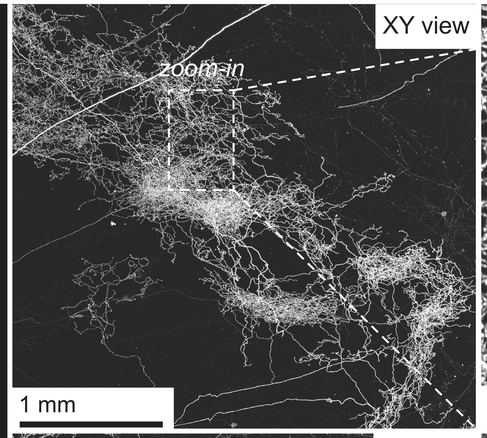


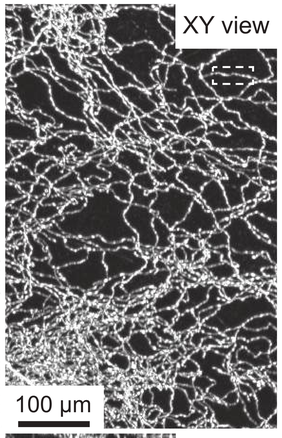


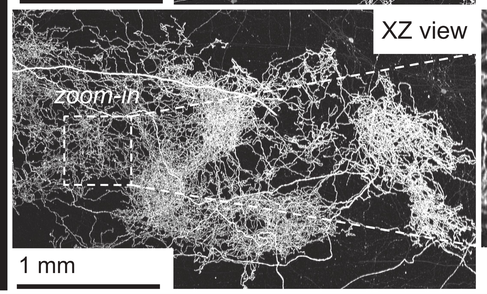
Granite base plate

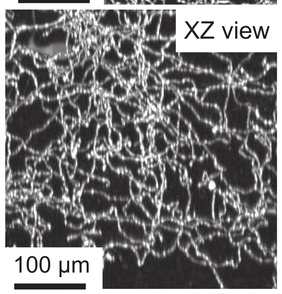












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NIH National Institutes of Health

BRAIN INITIATIVE